1. **Approach and tradeoffs**

Currently, each request runs two counts over ~1M rows contributing to the CPU hitting 85% at ~200 requests/min. I would move this computation from per-request to periodic refreshes.

**Freshness target:** Standing and percentile may lag by about ~5 minutes. The user's own CSR is always live.

**Design:** A Supabase cron job refreshes two materialized views every 5 minutes:

- **Score distribution**: (scope, score, users_at_or_above)storing cumulative eligible-user counts per CSR and scope (global/continent), plus total eligible users with CSR > 0. Eligibility and continent mapping are applied at refresh time.
- **User snapshot**: one row per eligible user holding their continent and the score they had at refresh time. Allows self-correction against live CSR.

Both views will have unique indexes for REFRESH MATERIALIZED VIEW CONCURRENTLY, allowing reads during refreshes. Both refreshes would then run within a single transaction, committing only if both succeed. This ensures readers see a consistent refresh generation.

**Request flow:**

1. Read the live CSR from AverageRanks (one-row lookup).
2. Find the smallest score strictly above the live CSR. Its cumulative count gives userAbove (or 0 if absent).
3. If the user's snapshot record belongs to the requested scope and its score exceeds their live CSR, subtract 1 to exclude their old self
4. Return Standing = usersAbove +1, percentile = usersAbove * 100 / total, and live CSR

**Tradeoffs:**

- Rankings lag other user's activities by approx. 5 min
- Floating-point CSR can yield nearly as many distinct scores as users, increasing refresh and index maintenance costs. In the real world I would benchmark this proposed implementation with a similar strategy that computes cumulative counts at fixed score boundaries (rank bucket population distribution table) plus an indexed in-bucket count. This trades smaller snapshots for more expensive reads.
- Near-ties at the top of the board can be briefly stale. If required, a live top-N path could be added.

The original code uses a global percentile denominator for continent scope and doesn't filter friends. I'd preserve both behaviors initially and flag them for correction.

2. **Rollout and verification**

- **Staging:** build the objects, unique index, and cron job on production-sized data, measuring refresh time and read latency. Confirm if avg latency < 200ms.

- **Correctness check:** verify the new path against an exact COUNT over the same snapshot, covering ties, score drops, continent membership, missing users, and empty scopes.
- **Shadow mode**: Run both implementations on sampled requests, return the old result, and log differences attributable to snapshot staleness
- **Ramp** 5% -> 25% -> 50% -> 100%, advancing only if latency and DB CPU are in thresholds. The flag provides instant rollback if thresholds are exceeded.
- **Cleanup:** Remove the old path after one clean release cycle

### 3. Post-launch monitoring and risks

Metrics: endpoint p50/p95 latency (target avg <200ms), DB CPU at peak (baseline 85%), snapshot age, refresh duration and failures, and sampled standing error versus the live count.

Risks**:**

- **Refresh fails silently and data goes stale:** Alert if snapshot age exceeds 15 minutes; retry, and fall back to the live query via the flag.
- **Refresh slows as users grow:** Alert when refresh duration exceeds ~20% of the interval (or 1 minute). Lengthen the interval, bucketed distributions, or move to a Redis sorted set.